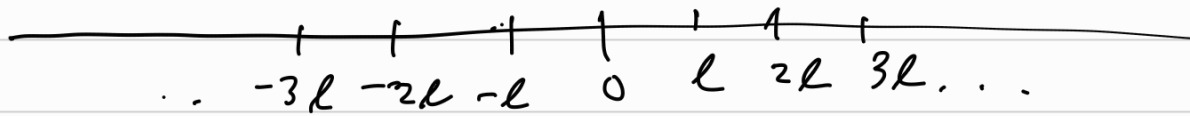


1D Random Walk:

Location of particle: $x = ml$, $-N \leq m \leq N$



after N total steps.

Want to determine the prob. $P_N(m)$ of finding the particle at the position $x = ml$ after N steps.



$$N = n_1 + n_2 \text{ (total steps)}$$

Net displacement:

$$m = n_1 - n_2$$

If we know n_1 and N then m is determined

$$m = n_1 - (N - n_1) = 2n_1 - N$$

$$m = 2n_2 - N$$

Claim 1) If N is odd, m must too be odd:

$$\rightarrow N = 2u + 1$$

$$m = 2n_1 - 2u - 1 + (1 - 1)$$

$$= 2(n_1 - u - 1) + 1$$

$$= 2\alpha + 1$$

Claim 2) If N is even, m is also even

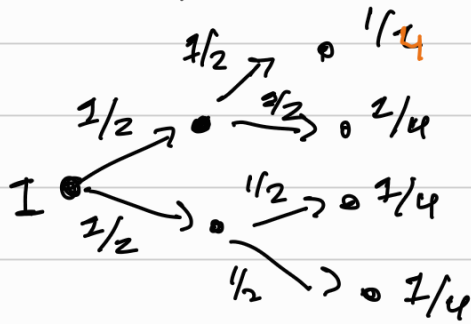
$$m = 2n_1 - N = 2n_1 - 2u = 2(n_1 - u) \checkmark$$

Each step is statistically independent of the previous step:

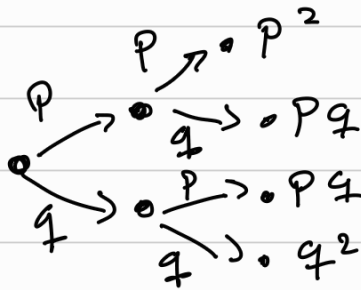
p : prob to the right

$q = 1 - p$: prob to the left

* Separate prob are multiplied:



no So each succession is splitting the odds.
 Say you're breaking a card into 4. Each
 break in succession are taken into account.



$$p^2 + 2pq + q^2 = 1$$

$$(p+q)^2 = 1 \quad \checkmark$$

=> Thus, after $n_1 + n_2 = N$ steps the
 probability of this is given by

$$\underbrace{p \dots p}_{n_1} \underbrace{q \dots q}_{n_2} = p^{n_1} q^{n_2}$$

• Must account for permutations:

1 2 ... N

1 | N : Can place N different objects there

2 | $N-1$: For each choice of the above
 we can choose $N-1$ ways to
 place here

⋮

N | 1

* Total of $N!$ different ways to place the objects

* Above assumes the items are distinguishable
 $\rightarrow n_i$ are indistinguishable so we have overcounted
 to rid ourselves of this redundancy we have

$$N! \rightarrow N! / n_1!$$

$$\rightarrow \text{Similarly, for } n_2, \quad N! / n_1! \rightarrow N! / n_1! n_2!$$

$$N! / n_1! n_2! = \frac{N!}{n_1! (N-n_1)!} = \binom{N}{n_1}$$

* This is the distinct number of ways in which N objects can be arranged given n_i type.

* Probability $W_N(n_1)$ of taking n_1 steps to the right and $n_2 = N - n_1$ to the left (in any order)

$$W_N(n_1) = \frac{N!}{n_1! n_2!} p^{n_1} q^{n_2}$$

ex: $N = 3$: $3 = n_1 + n_2$

$$m = 2n_1 - 3$$

$$n_1 = 0 \mid m = -3 \quad x = -3\ell$$

$$W_N(0) = q^3$$

$$n_1 = 1 \mid m = -1 \quad x = -\ell$$

$$W_N(1) = \frac{3!}{2!} p^1 q^2 = 3 p q^2$$

$$n_1 = 2 \mid m = 1 \quad x = \ell$$

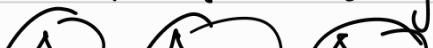
$$W_N(2) = 3 p^2 q$$

$$n_1 = 3 \mid m = 3$$

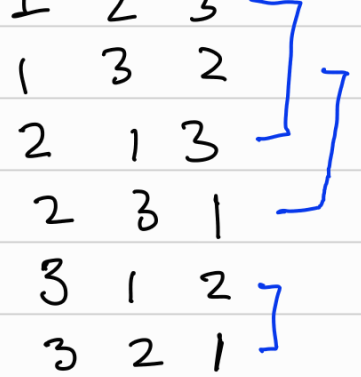
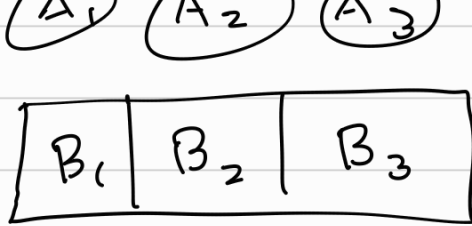
$$W_N(3) = p^3$$

$$x = 3\ell$$

ex: Showing how identical elements lead to identical configurations:



1 2 3



• Total of $3! = 3 \cdot 2 = 6$ possible can be made.

• Now, let " $A_1 = A_2$ " i.e. they are identical. then — indicates the redundancies: This leaves only 3 real possibilities

$$3! / 2! \cdot 1!$$

* Binomial distribution: $W_N(n,1) = \binom{N}{n_1} p^{n_1} q^{N-n_1}$

Binomial expansion: $(p+q)^N = \sum_{n=0}^N \frac{N!}{n!(N-n)!} p^n q^{N-n}$

proof:

$N=0$: $1 = 1 \checkmark$

$N=1$: $p+q = q + p^1 \checkmark$

$N=k$: Assume true for $k \in \mathbb{N}$, $k > 1$

$k+1$:

$$\begin{aligned} (p+q)^{k+1} &= (p+q)^k (p+q) \\ &= \sum_{n=0}^k \frac{k!}{n!(k-n)!} p^n q^{k-n} (p+q) \end{aligned}$$

$$= \sum_{n=0}^k \frac{k!}{n!(k-n)!} (p^{k+1} q^{k-n} + p^k q^{k-n+1})$$

proof involves using

$$\binom{k}{n} = \binom{k-1}{n} + \binom{k-1}{n-1}$$

$$\frac{(k-1)!}{n!(k-1-n)!} + \frac{(k-1)!}{(n-1)!(k-n)!}$$

$$= (k-1)! \left[\frac{(k-n)}{n!(k-n)!} + \frac{n}{n!(k-n)!} \right] = \frac{k!}{n!(k-n)!}$$

• Prob. that the particle is found at matter N steps is equivalent to $W_N(n_1)$

$$P_N(m) = W_N(n_1)$$

$$\begin{cases} N = n_1 + n_2 \\ M = n_1 - n_2 \end{cases} \Rightarrow \begin{cases} (N+M)^{\frac{1}{2}} = n_1 \\ (N-M)^{\frac{1}{2}} = n_2 \end{cases}$$

$$P_N(m) = \frac{N!}{[\frac{1}{2}(N+M)]! [\frac{1}{2}(N-M)]!} P^{\frac{1}{2}(N+M)} q^{\frac{1}{2}(N-M)}$$

* General discussion of mean value:

$$\bullet x \in \{x_1, \dots, x_m\} \quad M \in \mathbb{M}$$

each has respective probability $P(x_1), \dots, P(x_m)$

Def: The mean (average) value of x

$$\bar{x} = \langle x \rangle = \frac{\sum_{i=1}^m P(x_i) x_i}{\sum_{i=1}^m P(x_i)}$$

For any function $f: \{x_1, \dots, x_m\} \rightarrow ?$ which is a Ring homomorphism, the average is preserved

$$\begin{aligned} f(\bar{x}) &= \sum_{i=1}^m f\left(\left(\sum_{i=1}^m P(x_i) x_i\right)\right) \\ &= \left(\sum_{i=1}^m P(x_i)\right) f\left(\sum_{i=1}^m x_i\right) \end{aligned}$$

$$\bullet \text{ In general, } \sum_{i=1}^m P(x_i) = 1 \quad \begin{matrix} \text{(Normalization)} \\ \text{(Condition)} \end{matrix}$$

$$\langle f(x) \rangle = \sum_{i=1}^m P(x_i) f(x_i)$$

$$* \langle f(x) + g(x) \rangle = \langle f(x) \rangle + \langle g(x) \rangle$$

$$\langle c \cdot f(x) \rangle = c \langle f(x) \rangle$$

$\leadsto \langle x \rangle$ is a measure of the "middle" value of x .

→ Measure the difference in x from the middle $\bar{x}$

$$\Delta x = x - \bar{x}$$

ex: Take a set of letter grades with an average of $69 = \bar{x}$. Then

$$\Delta x = x - 69$$

$$(a) x = 80 : \Delta x = 11$$

$$(b) x = 60 : \Delta x = -9$$

• The average difference from $\bar{x}$ should be zero:

$$\langle \Delta x \rangle = \langle x - \bar{x} \rangle = \bar{x} - \bar{x} = 0$$

• Second moment of x about its mean:

(dispersion of x):

$$\langle (\Delta x)^2 \rangle \equiv \sum_{i=1}^M P(x_i) (x_i - \bar{x})^2 \geq 0$$

• Since Δx^2 is positive $\langle \Delta x^2 \rangle$ is positive

Only vanishes if $x_i = \bar{x} \forall i$

$\langle \Delta x^2 \rangle$: Measures the amount of scatter of values of the variable about its mean value.

$$\begin{aligned} \langle \Delta x^2 \rangle &= \langle x^2 - 2x\bar{x} + \bar{x}^2 \rangle \\ &= \langle x^2 \rangle - \langle x \rangle^2 \end{aligned}$$

$\therefore$

$$\overline{(\Delta x^2)} = \overline{x^2} - (\bar{x})^2$$

and so,

$$\langle \Delta x^2 \rangle \geq 0 \Rightarrow \langle x^2 \rangle \geq \langle x \rangle^2$$

• This says that the dispersion of x^2 is larger than the square of the dispersion of x

* Knowledge of $P(x)$ implies complete information about the actual distribution of values of the variable x .

→ Knowledge of only $\bar{x}$ and $(\Delta x)^2$ yields only partial knowledge.

Calculation of mean values for the random walk problem

- N total steps, n_1 to the right

$$W(n_1) = \frac{N!}{n_1! (N-n_1)!} p^{n_1} q^{N-n_1}$$

This is the probability of being at $x = ml$
 $m = n_1 - n_2$, or of going n_1 steps to the right.

- Claim: $\sum_{n_1=0}^N W(n_1) = 1$

Proof: $\sum_{n_1=0}^N \binom{N}{n_1} p^{n_1} q^{N-n_1} = (p+q)^N = 1^N = 1$

Q: What is the mean number of steps to the right $\bar{n}_1$?

$$\langle n_1 \rangle = \sum_{n_1=0}^N W(n_1) n_1 = \sum_{n_1=0}^N \binom{N}{n_1} p^{n_1} q^{N-n_1} n_1$$

- Assume p and q are separate variables
 $n_1 p^{n_1} = p \frac{\partial}{\partial p} (p^{n_1})$

$$\begin{aligned} \langle n_1 \rangle &= \sum_{n_1=0}^N \binom{N}{n_1} p \frac{\partial}{\partial p} (p^{n_1}) q^{N-n_1} = p \frac{\partial}{\partial p} \left[\sum_{n_1=0}^N \binom{N}{n_1} p^{n_1} q^{N-n_1} \right] \\ &= p \frac{\partial}{\partial p} (p+q)^N = Np (p+q)^{N-1} \end{aligned}$$

∴

$$\langle n_1 \rangle = Np (p+q)^{N-1}$$

- Small note: $\partial/\partial p \leftrightarrow \sum$ exchange is done since $\sum$ is finite & convergent.

- Now, with $p+q=1$,

$$\langle n_1 \rangle = Np$$

* Logically, this could be seen as follows:

Since p is the prob. of making a step towards n_1 , the mean number of right steps in a total of N steps is Np .

$$\langle n_2 \rangle = Nq$$

$$\langle n_1 \rangle + \langle n_2 \rangle = N$$

• Displacement: $m = n_1 - n_2$

$$\langle m \rangle = pN - qN = N(p - q)$$

* Calculation of the dispersion: $\overline{\Delta n_1^2}$

$$\langle \Delta n_1^2 \rangle = \langle n_1^2 \rangle - \langle n_1 \rangle^2$$

• Already know $\langle n_1 \rangle^2 = (Np)^2$

→ Find $\langle n_1^2 \rangle$: $n_1^2 = f(n_1)$

$$\langle f(n_1) \rangle = \sum_{n_1=0}^N W(n_1) f(n_1)$$

$$\Rightarrow \langle n_1^2 \rangle = \sum_{n_1=0}^N \binom{N}{n_1} p^{n_1} q^{N-n_1} n_1^2$$

• $n_1^2 p^{n_1} = n_1 (n_1 p^{n_1}) = n_1 \left(p \frac{\partial}{\partial p} \right) (p^{n_1})$

$$= \left(p \frac{\partial}{\partial p} \right) (n_1 p^{n_1}) = \left(p \frac{\partial}{\partial p} \right) \left(p \frac{\partial}{\partial p} \right) (p^{n_1})$$

$$n_1^2 p^{n_1} = \left(p \frac{\partial}{\partial p} \right)^2 (p^{n_1})$$

$$\langle n_1^2 \rangle = \sum_{n_1=0}^N \binom{N}{n_1} \left[p \frac{\partial}{\partial p} \right]^2 (p^{n_1} q^{N-n_1})$$

$$= \left[p \frac{\partial}{\partial p} \right]^2 \sum_{n_1=0}^N \binom{N}{n_1} p^{n_1} q^{N-n_1}$$

$$= \left(p \frac{\partial}{\partial p} \right)^2 (p+q)^N$$

$$= p \frac{\partial}{\partial p} (Np (p+q)^{N-1})$$

$$= p [N (p+q)^{N-1} + N(N-1)p (p+q)^{N-2}]$$

using $(p+q) = 1$

$$\langle n_1^2 \rangle = p [N + N(N-1)p]$$

$$= pN + N^2 p^2 - Np^2$$

$$= \langle n_1 \rangle^2 + Np(1-p)$$

$$\langle n_1^2 \rangle = \langle n_1 \rangle^2 + Npq$$

Plugging this back into our result yields

$$\overline{\Delta n_1^2} = \langle n_1^2 \rangle - \langle n_1 \rangle^2$$

$$\langle \Delta n_1^2 \rangle = Npq$$

- Quadratic in displacement, so we could have a linear measurement

* Root-Mean-Square (RMS): $\Delta^* u \equiv \sqrt{\langle \Delta^2 \rangle}$
 $\leadsto$ linear measure of the width of the range over which n_1 is distributed

Relative width: $\frac{\Delta^* n_1}{n_1} = \frac{\sqrt{Npq}}{Np} = \sqrt{\frac{q}{Np}}$

- Note as N increases, $\langle n_1 \rangle$ increases as $\sim N$, while $\Delta^* n_1$ (the width) increases as $\sim N^{1/2}$. Relative width is decreasing as $\sim N^{-1/2}$

* Dispersion of m : $m = n_1 - n_2 = 2n_1 - N$
 $\Delta m = \langle m^2 \rangle - \langle m \rangle^2$
 $= \langle 4n_1^2 - 4n_1 N + N^2 \rangle - \langle 2n_1 \rangle^2 - \langle N \rangle^2$
 $\quad + \langle 4n_1 N \rangle$
 $= 4\langle n_1^2 \rangle - 4N\langle n_1 \rangle + N^2 - 4\langle n_1 \rangle^2 - N^2 + 4N\langle n_1 \rangle$

$\Delta m = 4Npq$

or

$\Delta m = 4\Delta u$

Prob. dist. for large N :

- Approximate $W(n_1)$ for large N :
 $\leadsto$ Consider the region near the maximum of W where $\tilde{n}_1$ is also large:

$|W(\tilde{n}_1 + 1) - W(\tilde{n}_1)| \ll W(\tilde{n}_1)$

To a good approx. $N \rightarrow \infty$ and so

$W(n_1) = \sum_{n_1} () \rightarrow \int () dn_1$

$$\tilde{n} \equiv \max(W(n_1))$$

$$\rightarrow \left. \frac{dW}{dn_1} \right|_{n_1=\tilde{n}} = 0 \quad \text{or equivalently} \quad \left. \frac{d \ln(W)}{dn_1} \right|_{n_1=\tilde{n}} = 0$$

$\rightarrow$ to investigate the behavior near $\tilde{n}_1$ let

$$n_1 \equiv \tilde{n}_1 + \mathcal{R}$$

Expand in a Taylor series in $\tilde{n}_1$. Do $\ln(W)$ instead of W since $\ln(W)$ much more slowly varying f^n than W

Ex: Want to find an approx. expression, valid for $y \ll 1$

$$f = (1+y)^{-N}$$

$$N \gg 1 \rightarrow \sum_{n=0}^{\infty} f^{(n)}(0) y^n / n!$$

$$= 1 - Ny + \frac{1}{2} N(N+1) y^2 - \dots$$

Since N is large, $Ny \geq 1$ and so the above no longer converges

• Get around this by taking the $\ln(\cdot)$ first

$$\ln(f) = -N \ln(1+y)$$

$$\ln(f) = -N \left[0 + y - \frac{1}{2} y^2 + \dots \right]$$

$$= -N \left(y - \frac{1}{2} y^2 + \dots \right)$$

$$\therefore f = \exp \left[-N \left(y - \frac{1}{2} y^2 + \dots \right) \right]$$

Thus, we have, approximately, a Gaussian dist. as long as $y \lesssim 1$

Now, we have: $n_1 = \tilde{n}_1 + \mathcal{R}$

$$\ln(W(n_1)) = \sum_{n=0}^{\infty} \frac{\partial^n}{\partial n_1^n} \left(\ln(W(n_1)) \right) \Big|_{\tilde{n}_1} \mathcal{R}^n$$

$\left\{ \begin{array}{l} z = x + \epsilon \\ f(z) = \sum_{n=0}^{\infty} f^{(n)}(x) / n! \epsilon^n \end{array} \right\}$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(\tilde{n}_1)}{n!} (x - \tilde{n}_1)^n \quad \left| \quad B_k \equiv \frac{\partial^k}{\partial n_1^k} (\ln(W(n_1))) \right|_{n_1 = \tilde{n}_1}$$

$$= \ln(W(\tilde{n}_1)) + B_1 \mathcal{R} + \frac{1}{2} B_2 \mathcal{R}^2 + \dots$$

- Since we are expanding about a maximum:

$$B_1 = \frac{\partial}{\partial n_1} (\ln(W(n_1))) = 0$$

- If $B_1 |_{\tilde{n}_1}$ is a maximum, then $B_2 < 0$
hence $\frac{1}{2} B_2 \mathcal{R}^2 < 0$. Explicitly,

$$B_2 = -|B_2|$$

$$\ln(W(n_1)) = \ln(W(\tilde{n}_1)) - \frac{1}{2} |B_2| \mathcal{R}^2 + \dots$$

Let $\tilde{W} \equiv W(\tilde{n}_1)$, then

$$W(n_1) = \exp(\ln(\tilde{W}) - \frac{1}{2} |B_2| \mathcal{R}^2 + \frac{1}{6} B_3 \mathcal{R}^3 + \dots)$$

$$\Rightarrow W(n_1) = \tilde{W} \exp(-\frac{1}{2} |B_2| \mathcal{R}^2 + \frac{1}{6} B_3 \mathcal{R}^3 + \dots)$$

For extremely small $\mathcal{R}$ we have, to $\mathcal{O}(\mathcal{R}^2)$

$$W(n_1) = \tilde{W} e^{-\frac{1}{2} |B_2| \mathcal{R}^2}$$

- We now consider the explicit form of W

$$\begin{aligned} \ln(W(n_1)) &= \ln\left(\frac{N!}{n_1!(N-n_1)!} p^{n_1} q^{N-n_1}\right) \\ &= \ln\left(\binom{N}{n_1}\right) + n_1 \ln(p) + (N-n_1) \ln(q) \\ &= \ln(N!) - \ln(n_1!) - \ln((N-n_1)!) + n_1 \ln(p) \\ &\quad + (N-n_1) \ln(q) \end{aligned}$$

- If $n_1 \gg 1$, then $\ln(n_1!)$ can be considered an almost continuous fn

$$\begin{aligned} \frac{d \ln(n_1!)}{dn} &\approx \lim_{\Delta n \rightarrow 0} \left(\frac{\ln((n+1)!) - \ln(n!)}{((n+1) - n)} \right) \\ &= \lim_{\Delta n \rightarrow 0} \left(\ln\left(\frac{(n+1)!}{n!}\right) \right) = \ln(n+1) \end{aligned}$$

* If we take $n \gg 1$ then

$$\ln(n+1) \rightarrow \ln(n)$$

$$\frac{d \ln(n!)}{dn} \approx \ln(n)$$

• Thus: $\frac{d \ln(w(n_1))}{dn_1} = -\ln(n_1) + \ln(N-n_1) + \ln(p) - \ln(q)$

Set to zero to find the minimum:

$$0 = -\ln(n_1) + \ln(N-n_1) + \ln(p) - \ln(q)$$

$$\Rightarrow 0 = \ln\left(\frac{N-n_1}{n_1} \cdot \frac{p}{q}\right)$$

$$1 = \left(\frac{N-n_1}{n_1}\right) \left(\frac{p}{q}\right) \Rightarrow \hat{n}_1 q = pN - \hat{n}_1 p$$

$$\hat{n}_1 = pN$$

which is exactly $\langle n_1 \rangle = pN$

$$\frac{d^2 \ln(w)}{dn_1^2} = -\frac{1}{n_1} - \frac{1}{N-n_1}$$

$$\frac{d^2 \ln(w)}{dn_1^2} \Big|_{n_1=\hat{n}_1} = -\frac{1}{Np} - \frac{1}{N-Np} = -\frac{2}{N} \left(\frac{1}{p} + \frac{1}{q}\right)$$

$$\therefore B_2 = -\frac{1}{Npq}$$

* Arguments for ignoring higher order terms $\mathcal{O}(n^3)$:

$$B_3 \Big|_{\hat{n}_1} = \frac{2}{\hat{n}_1} \left(-\frac{1}{n_1} - \frac{1}{N-n_1}\right) = \frac{1}{\hat{n}_1^2} - \frac{2}{(N-\hat{n}_1)^2}$$

$$= \frac{1}{N^2 p^2} - \frac{2}{N^2 (1-p)^2} = \frac{1}{N^2} \left(\frac{1}{p^2} - \frac{1}{q^2}\right)$$

$$\therefore |B_3| = \left(\frac{1}{N^2 p^2 q^2}\right) |q^2 - p^2|$$

• Claim $|q^2 - p^2| < 1$

Proof: $p + q = 1$

$$\therefore p^2 + 2pq + q^2 = 1$$

$$p^2 - q^2 = 1 - 2q^2 - 2pq$$

$$\therefore |p^2 - q^2| = |1 - 2q^2 - 2pq|$$

$$< 1$$

Hence,

$$|B_3| = |B_2| |q^2 - p^2| < |B_2|$$

• Similarly $|B_4| < \frac{1}{Npq} |B_2|$

• B_n is smaller than $n^k / (Npq)^{k-1}$

↳ Ignore B_3 for N small enough that
 $N \ll Npq$

• Determine $\tilde{W}$ via the normalization of W

$$\sum_{n_1=0}^N W(n_1) \approx \int W(n_1) dn_1 = \int_{-\infty}^{\infty} W(\tilde{n} + \tilde{n}) d\tilde{n} = 1$$

$$= \int_{-\infty}^{\infty} \tilde{W} e^{-\frac{1}{2}|B_2|\tilde{n}^2} d\tilde{n} = \tilde{W} \sqrt{\frac{\pi}{(\frac{1}{2}|B_2|)}} = \tilde{W} \sqrt{\frac{2\pi}{|B_2|}} = 1$$

$$\therefore \tilde{W} = \sqrt{\frac{|B_2|}{2\pi}} \quad |B_2| = \frac{1}{Npq}$$

Thus, $W(n_1) = \tilde{W} e^{-\frac{1}{2}|B_2|n_1^2}$ is given by ($n_1 = n_1 - \tilde{n}$)

$$W(n_1) = (2\pi Npq)^{-1/2} \exp\left[-\frac{(n_1 - Np)^2}{2 Npq}\right]$$

$$\langle n_1 \rangle = Np, \quad \langle \Delta n_1^2 \rangle = Npq$$

$$W(n_1) = (2\pi \overline{\Delta n_1^2})^{-1/2} \exp\left[-\frac{(n_1 - \bar{n}_1)^2}{2 \overline{\Delta n_1^2}}\right]$$

Gaussian Probability distributions:

• $P(m)$: prob. that in a large number of steps the net displacement is m .

$$n_1 = \frac{1}{2}(N+m)$$

$$P(m) = W\left(\frac{N+m}{2}\right) = [2\pi Npq]^{-1/2} \exp\left[-\frac{[m - N(p-q)]^2}{8Npq}\right]$$

$$m = 2n_1 - N, \quad \text{so } m \in \mathbb{Z}$$

$$\Delta m = m - \langle m \rangle = 2\Delta n_1 = 2(n_1 - Np)$$

$$= 2\left(\frac{1}{2}(N+m) - Np\right)$$

$$= N+m - 2Np = m + N(1-p-p)$$

$$= m + N(q-p)$$

$$\Delta m = m - N(p-q) \quad \text{at } n_1 = \tilde{n} = Np$$

$$\hookrightarrow \text{Perhaps } \langle m \rangle \leadsto \int W(n_1) m$$

• Book states $\Delta m = 2$.

* Express, $x = mL$

• dk contains dx / 20 values of m

$$\leadsto \underline{x = ml} \Rightarrow dx = dl$$

however it must be $dx/2l = dm$

- Why? $x = \tilde{m}l$ but $\tilde{m} = 2m$ and so
for $m \in \mathbb{Z} \rightarrow$ discrete.

$$dx = 2l dm$$

$$P(x) dx = P(m) \frac{dx}{2l}$$

i. e. $P(x) = \frac{P(m)}{2l}$

* Define:

$$\mu = (p-q) \cdot Nl$$

$$\sigma = 2 \sqrt{Npq} l$$

$$P(x) dx = (2\pi Npq)^{-1/2} \exp\left(-\frac{(m - N(p-q))^2}{8Npq}\right) dx$$

$$\Rightarrow (2\pi Npq)^{-1/2} = \frac{1}{\sqrt{2\pi}} \left(\frac{1}{2l}\right) \exp\left(-\frac{(ml - N(p-q)l)^2}{8l^2 Npq}\right)$$

$$\exp\left[-\frac{(x-\mu)^2}{8l^2 Npq}\right] = \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$

$$\therefore P(x) dx = \frac{1}{\sqrt{2\pi} \sigma} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$

Normalisation: $\int P(x) dx = \frac{1}{\sqrt{2\pi} \sigma} \int_{-\infty}^{\infty} dx e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

$$\int dy e^{-y^2/2\sigma^2} = \sqrt{\frac{\pi}{\left(\frac{1}{2\sigma^2}\right)}} = \sqrt{2\pi \sigma^2}$$

$$\therefore \int_{-\infty}^{\infty} dx e^{-\frac{(x-\mu)^2}{2\sigma^2}} \frac{1}{\sqrt{2\pi} \sigma} = \frac{\sqrt{2\pi \sigma^2}}{\sqrt{2\pi} \sigma} = 1$$

i. e. it is normalized.

* Mean value of x

$$\langle x \rangle = \int_{-\infty}^{\infty} dx x e^{-\frac{(x-\mu)^2}{2\sigma^2}} \frac{1}{\sqrt{2\pi} \sigma}$$

$$= \int_{-\infty}^{\infty} dy (\cancel{x} + \mu) e^{-y^2/2\sigma^2}$$

$$= \int_{-\infty}^{\infty} dy \, y \, e^{-y^2/2\sigma^2} + \mu \int_{-\infty}^{\infty} dy \, e^{-y^2/2\sigma^2}$$

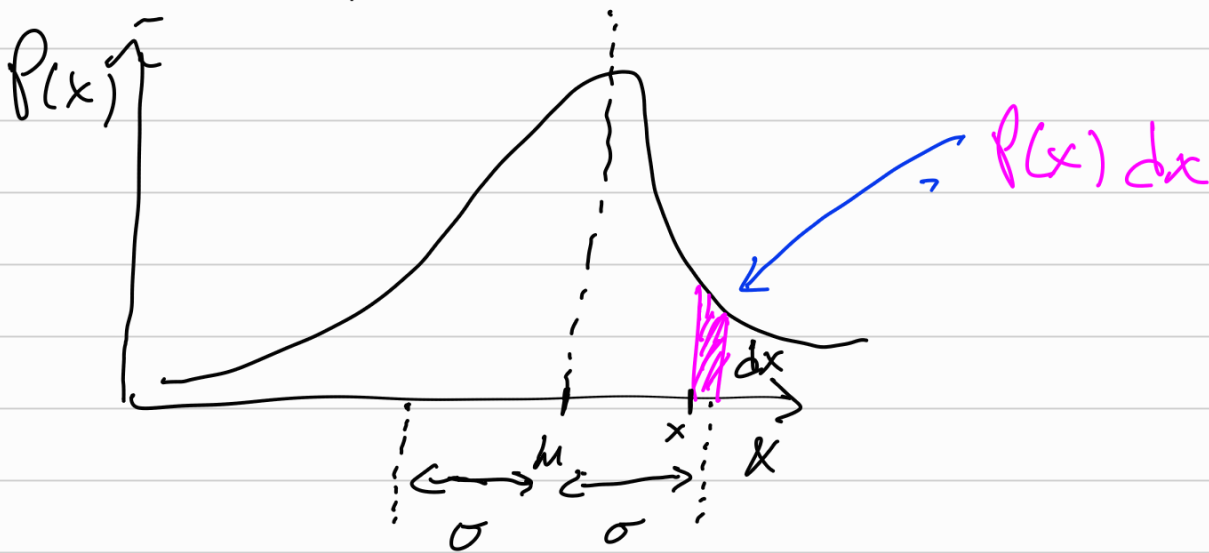
$$= \int_{-\infty}^{\infty} dy \, y \, e^{-y^2/2\sigma^2} + \mu$$

• $y \rightarrow -y$: $\int_{-\infty}^{\infty} dy \, (-y) \, e^{-y^2/2\sigma^2}$
 i.e. odd fⁿ over a symmetric interval
 means $\int_{-\infty}^{\infty} dy \, y \, e^{-y^2/2\sigma^2} = 0$

∴

$$\langle x \rangle = \mu$$

$P(x)$ is symmetrical about μ :



$$\langle (x-\mu)^2 \rangle = \int_{-\infty}^{\infty} dx \, (x-\mu)^2 \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$\int_{-\infty}^{\infty} dy \, e^{-\beta y^2} = \sqrt{\frac{\pi}{\beta}}$$

$$\therefore \frac{\partial}{\partial \beta} \left(\int_{-\infty}^{\infty} dy \, e^{-\beta y^2} \right) = \int_{-\infty}^{\infty} dy \, (-y^2) e^{-\beta y^2} = - \int_{-\infty}^{\infty} dy \, y^2 e^{-\beta y^2}$$

$$= - \frac{1}{2} \frac{\sqrt{\pi}}{\beta^{3/2}}$$

$$\therefore \int_{-\infty}^{\infty} dy \, y^2 e^{-\beta y^2} = \frac{1}{2} \sqrt{\frac{\pi}{\beta^3}}$$

$$\therefore \int_{-\infty}^{\infty} dx \, (x-\mu)^2 e^{-\frac{(x-\mu)^2}{2\sigma^2}} \frac{1}{\sqrt{2\pi}\sigma} = \frac{1}{\sqrt{2\pi}\sigma} \left(\frac{1}{2} \sqrt{\pi (2\sigma^2)^3} \right)$$

$$= \left(\frac{1}{2^{3/2}} \right) \frac{1}{\sigma} = \sigma^2$$

$$\therefore \boxed{\langle (x - \mu)^2 \rangle = \sigma^2} \quad \text{or} \quad \boxed{\langle (\Delta x)^2 \rangle = \sigma^2}$$

σ : Root-Mean-Square deviation of x from the mean value.

$$\sigma = \sqrt{\langle (\Delta x)^2 \rangle}$$

$$\langle x \rangle = (p - q)Nl$$

$$\langle (\Delta x)^2 \rangle = 4Npq l^2$$

This is what was Already found!

Found $\langle m \rangle = N(p - q)$

$$\langle \Delta m^2 \rangle = 4Npq$$

$$\therefore \langle x \rangle = \langle m l \rangle = Nl(p - q)$$

$$\langle (\Delta x)^2 \rangle = \langle (\Delta m)^2 l^2 \rangle = \langle \Delta m^2 \rangle l^2 = l^2 4Npq$$

* All this for a 1D Random walk.

General Random Walk ($D > 1$):

* Use general prob. theory and not build off the 1D model.

2D:
$$\begin{array}{ll} r_i & i = 1, \dots, M \\ n_j & j = 1, \dots, N \end{array}$$

$\Rightarrow P(r_i, n_j)$: prob. that r assumes values r_i and n assumes the values of n_j .

$$\sum_{i=1}^M \sum_{j=1}^N P(r_i, n_j) = 1$$

$$P_r(r_i) = \sum_{j=1}^N P(r_i, n_j)$$

$$P_n(n_j) = \sum_{i=1}^M P(r_i, n_j)$$

* Statistical Independence: $P(x_i, y_j) = P(x_i) P(y_j)$
 i.e. uncorrelated. Means they are statistically independent.

Let $F(x, y)$ be a f.h of (x, y) and so the mean value is given by

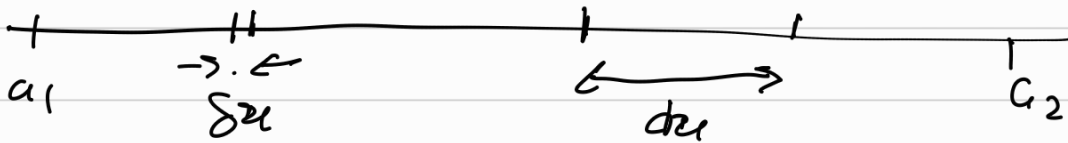
$$\langle F(x, y) \rangle = \sum_{i=1}^M \sum_{j=1}^N P(x_i, y_j) F(x_i, y_j)$$

Suppose $f = f(x)$, then

$$\begin{aligned} \langle f(x) \rangle &= \sum_{i=1}^M \sum_{j=1}^N P(x_i, y_j) f(x_i) = \sum_{i=1}^M \left(\sum_{j=1}^N P(x_i, y_j) \right) f(x_i) \\ &= \sum_{i=1}^M P_x(x_i) f(x_i) \end{aligned}$$

• $\langle F + G \rangle = \langle F \rangle + \langle G \rangle$

• $a_1 < x < a_2$



→ Prob. of being within dx
 $f(x) dx$

• Divide a_1 and a_2 in N smaller Δx
 $a_2 - a_1 = N \Delta x$
 or $\Delta x = (a_2 - a_1) / N$

• Label each subdiv. by an i

• Each value of x on Δx_i is labelled by x_i w/ prob within Δx_i given by $P(x_i)$.

$$P(x) = P(x) \Delta x$$

→ $\left(\frac{P(x)}{\Delta x} \right) dx = f(x) dx$

P: Discrete
P: Continuous

$$\sum_i P(u_i) = 1 \Leftrightarrow \int_{a_1}^{a_2} P(u) du = 1$$

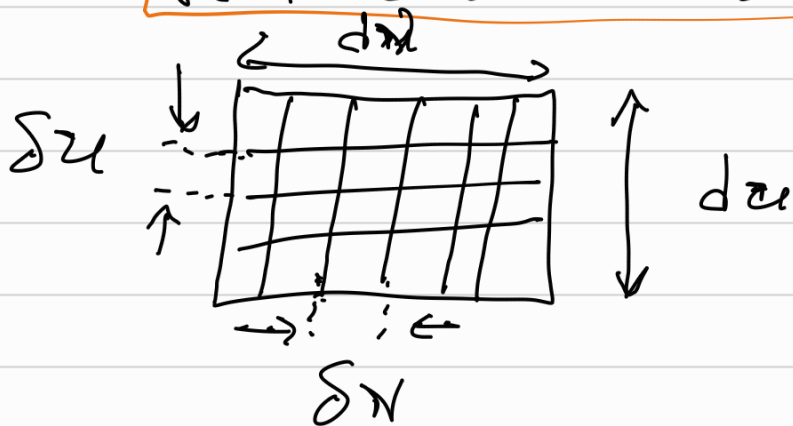
$$\langle f(u) \rangle_P = \sum_i P(u_i) f(u_i)$$

$$\langle f(u) \rangle_c = \int_{a_1}^{a_2} du P(u) f(u)$$

* 2-variables: (u, v)

Prob: $P(u, v) du dv$

$$P(u, v) du dv = P(u, v) \frac{du}{\delta u} \frac{dv}{\delta v}$$



- $du/\delta u$: # of divisions in du
 $dv/\delta v$: # of divisions in dv

$$\int_{a_1}^{a_2} \int_{b_1}^{b_2} du dv P(u, v) = 1$$

$$\langle F(u, v) \rangle = \int_{a_1}^{a_2} \int_{b_1}^{b_2} du dv P(u, v) F(u, v)$$